SOLUTIONS - Exercise - Week 3: MATH1061/7861

1. **(3 marks)** Is the following statement true or false? Justify your answer.

$$(p \wedge q) \to r \equiv (\sim p) \to (q \to r).$$

The statement is false.

Consider the valuation with p false, q true and r false. Looking at the left-hand side, $p \wedge q$ is false, so $(p \wedge q) \to r$ is true. However, looking at the right-hand side, we see that $q \to r$ is false, so $(\sim p) \to (q \to r)$ is false. Thus the two sides of the statement are not the same, and hence it is false.

MARKING:
1 mark for mentioning that the statement is false.
2 marks for justification (if they use laws of logical equivalence, they still must give a valuation to show the sides are not equivalent).
Using a truth-table is also correct justification.

2. **(3 marks)** Is the following argument valid? Justify your answer.

$$\begin{aligned} & p \to q \\ & p \vee r \\ & p \vee \sim r \\ & \therefore \sim q \end{aligned}$$

The argument is invalid.

Consider the valuation with p, q and r all true. Then the premises are true, as $p \to q$ is true, $p \vee r$ is true, and $p \vee \sim r$ is true. However, the conclusion $\sim q$ is false. Thus the argument is invalid.

MARKING:
1 mark for mentioning that the argument is invalid.
2 marks for justification (they could also use a truth table; or they could also show that the same premises imply q, and hence this argument is false).

3. **(4 marks)** For each of the following statements, write the negation. Determine whether the original statement or the negation is true. Justify your answer.

(a) $\forall x \in \mathbb{R},\ (x^2 \geq x) \rightarrow (x \geq 0)$.

Negation: $\exists x \in \mathbb{R}$ such that $(x^2 \geq x) \wedge (x < 0)$

The original statement is false.

For example, consider $x = -1$. Then $x^2 = (-1)^2 = 1 \geq -1$ and $x < 0$.

MARKING:
1 mark for stating the negation correctly.
1 mark for justification that the original statement is false.
Comment (with no deductions) if students do not use "such that" (or "s.t." etc) after "$\exists x \in \mathbb{R}$".

(b) $\forall x, y \in \mathbb{R}$, if $x + y \in \mathbb{Z}$ and $x - y \in \mathbb{Z}$ then $x \in \mathbb{Z}$.

Negation: $\exists x, y \in \mathbb{R}$ such that $x + y \in \mathbb{Z}$ and $x - y \in \mathbb{Z}$ and $x \notin \mathbb{Z}$.

The original statement is false.

For example, consider $x = 1/2$ and $y = 1/2$. Then $x + y = 1$ is an integer and $x - y = 0$ is an integer, but x is not an integer.

MARKING:
1 mark for stating the negation correctly.
1 mark for justification that the original statement is false.
Comment (with no deductions) if students do not use "such that" (or "s.t." etc) after "$\exists x \in \mathbb{R}$".